Fiscal Year:	FY 2024	**Task Last Updated:**	FY 05/07/2026
PI Name:	Hirzallah, Mohammad M.D.		
Project Title:	The Effect of Negative Pressure Ventilation on Intracranial Pressure (NPV ICP)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Catalyst
Start Date:	01/01/2024	**End Date:**	06/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-CAT0005		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Negative pressure ventilation (NPV) can be a Spaceflight Associated Neuro-ocular Syndrome (SANS) countermeasure by reducing intracranial pressure (ICP) and improve cerebral venous outflow. We will study the effects of NPV on invasive ICP and jugular vein area in patients with an invasive intracranial pressure monitor. This will be accomplished by measuring ICP and IJVA before the application of NPV and during NPV. Hypothesis: We propose using negative pressure ventilation (NPV) to create negative intrathoracic pressure and subsequently reduce ICP and improve cerebral venous outflow without the negative airflow restriction effects of impedance threshold devices (ITD). Methods: We will study the effects of NPV on invasive ICP and IJVA in patients with an invasive intracranial pressure monitor. This will be accomplished by measuring ICP and IJVA before the application of NPV and during NPV. Specific Aims: to determine if the application of NPV in patients with		

	neurological injuries and invasive ICP monitor: (1) reduces ICP; (2) reduces cerebral venous congestion as measured by internal jugular vein (IJV) cross sectional area (IJVA).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	NPV has potential as a SANS countermeasure that builds on the advantages of previously investigated mechanical countermeasures (ITD and lower body negative pressure/LBNP) while adding the advantages of lightweight, comfort, portability, dual purpose, and ease of use. This no more than minimal risk study has the potential deliverable of validating NPV as an easy-to-use, and effective SANS and cerebral venous congestion mechanical countermeasure.
Task Progress:	New Project for FY2024
Bibliography Type:	Description: (Last Updated:)